

Lóránt Nagy, PhD

- Budapest, Hungary
- nagy_lorant@phd.ceu.edu
- lorant-nagy.github.io

Mathematician with a strong command of probability, stochastic modelling, mathematical finance and modern statistical and machine-learning methods, with applied experience in capital-markets modelling, energy forecasting and deep learning.

Professional Experience

Eötvös Loránd University, ELTE AI Research Group **2023–Present**
Research Assistant

- Research deep-learning methods for statistical inference in stochastic-process and long-memory time-series models.
- Conducted an interpretability study on neural networks applied to time-series data: investigated which characteristics of the input drive the networks' inference.
- Led the technical development of reinforcement-learning methods for automated trading in stylized market environments, driving the project from problem formulation through model development to experimental validation.

Alfréd Rényi Institute of Mathematics **2019–Present**
Research Fellow

- Investigate alternative stochastic dynamics for diffusion-based generative models.
- Conduct research in mathematical finance, stochastic control, market frictions, and optimal investment.

Artificial Intelligence National Laboratory **2024**
ML Researcher

- Collaborated with ALTEO on intraday electricity-price forecasting.
- Developed and evaluated machine-learning regression models, with a focus on predictive reliability.

Sysler Kft. **2023**
Research Consultant

- Prototyped a Bayesian late-fusion framework combining outputs from independent medical-diagnostic models to infer disease progression from empirical patient trajectories.

Morgan Stanley **2017**
Interest Rate Modelling Intern

- Reviewed financial-product documentation, checking model assumptions, calculations, pricing results, and internal consistency.
- Reverse-engineered parts of a legacy pricing system written in A+ to understand existing model and pricing logic.

Technical Skills

Programming and development tools: Python, PyTorch, scikit-learn, NumPy, SciPy, Git, Docker, GitHub Actions, Wolfram Mathematica, TensorBoard, Optuna

Quantitative methods: stochastic modelling, simulation, optimization, statistical analysis, mathematical finance, statistical learning, time-series forecasting, information theory

Cloud and deployment: Azure Static Web Apps, Azure Container Apps, containerized deployment, basic CI/CD workflows

Machine learning and deep learning: generative models, large language model foundations, neural networks, variational inference, reinforcement learning, late fusion, Bayesian filtering, tree-based methods, dimensionality reduction, clustering

Education

Central European University <i>PhD in Financial Mathematics</i>	2018–2022
Budapest University of Technology and Economics <i>MSc in Applied Mathematics</i>	2015–2018

Selected Publications

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- 2026** Nagy, L. and Lovas, A.
Ornstein–Uhlenbeck Is Hard to Beat, Yet Superlinear Drift Ships Lower Transport Costs.
Preprint, arXiv:.
- 2026** Rásonyi, M. and Nagy, L.
Optimal investment with transient price impact and stochastic spread.
Preprint, arXiv:2511.12093.
- 2024** Boros, D., Csanády, B., Ivkovic, I., Nagy, L., Lukács, A., and Márkus, L.
Deep learning the Hurst parameter of linear fractional processes and assessing its reliability.
Quality and Reliability Engineering International, 40, 4228–4246.
- 2022** Guasoni, P., Nagy, L., Rásonyi, M.
Young, timid, and risk takers.
Mathematical Finance. 31: 1332–1356.
- 2021** Rásonyi, M. and Nagy, L.
Optimal long-term investment in illiquid markets when prices have negative memory.
Electronic Communications in Probability, 26.

Selected Talks

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- 2026** **Bologna** **XIII Bachelier World Congress**
On the utility problem in a market where price impact is transient
- 2025** **Vienna** **Congress on Mathematical Finance, TU Wien**
Non-convex optimal investment with transient price impact
- 2024** **Szeged** **Conference on Analysis and Applied Mathematics, University of Szeged**
Optimal execution in superlinearly mean-reverting environments
- 2023** **Budapest** **ELTE–Rényi AI Workshop, Eötvös Loránd University**
Risk-averse trading
- 2022** **Ann Arbor** **Financial Mathematics Seminar, University of Michigan**
Young, Timid, and Risk Takers

Additional Experience and Distinctions

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- 2025** Intermediate-level Qualification in Copyright Law, Hungarian Intellectual Property Office.
- 2022** Award for Advanced Doctoral Students.
- 2019** Instructor, *Fundamentals of Stochastic Analysis*, Central European University.
- 2016** Second Place, TDK competition, *Examples in the theory of continuous trading with friction.*